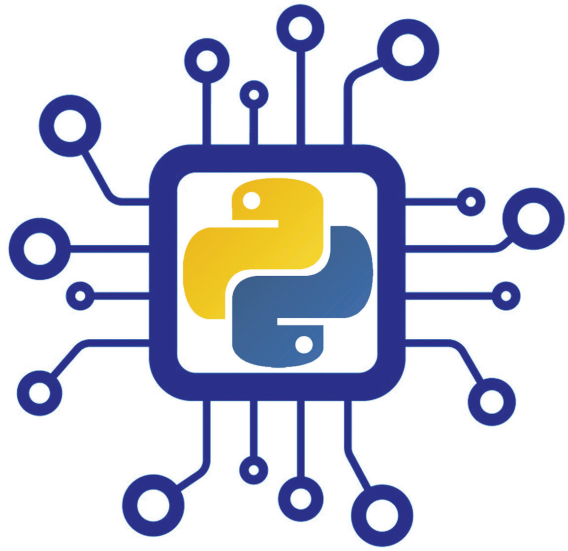
Python: The Super Champ For Machine Learning

Machine Learning (ML) is a technology used to make any software process improve through experience with the help of algorithms. It is a part of artificial intelligence (AI) and uses training data to improve decisions and predictions, without the need for any explicit programming. ML was first conceived by Arthur Samuel of IBM in 1952. This article highlights Python as the best language for ML



Debojit Acharjee

In business, ML is used to resolve various problems and the process is referred to as predictive analysis. To run any static process, it's possible to program a computer using specific algorithms. However, the computer must modify its program and improve the algorithm to handle dynamic processes in real time. Arthur Samuel of IBM created the world's first ML program to run checkers video games on IBM 701 computers in 1952.

There are mainly two objectives of ML—one is to classify data models, which have been developed using I/O data, and the other is to predict future outputs based on that data. These two objectives can be achieved using various learning algorithms, and also with the help of a powerful programming language like Python.

TYPES OF LEARNING APPROACHES

In ML, an agent can be made intelligent using various learning approaches and algorithms. In such a learning process, ML has to go through three stages—training, validation, and testing. This is done in order to process the input data and modify the algorithm for improving prediction and performance. There are various types of learning approaches available, but the three main traditionally used approaches are supervised, unsupervised, and reinforcement learning.

Supervised learning

As the name suggests, this learning approach involves training the agent by a supervisor, which is usually the user. Here, the training is given to the agent using a set of training data, as examples or data models. The data contains a label and a feature. The set of data also contains the mapping of the input and output data based on the data model. So, when any new data is fed into the agent as input data, it also generates new output data based on the data model. A simple example of a data model would be a table containing rows and columns, which has data about various laptops with different specifications and price tags. So, by looking at the table one can understand which laptops are expensive and which are cheap. If this model is used for ML training, then the agent would be able to predict if a laptop with higher CPU speed will cost more or less.

The accuracy and method of processing the data depends on the algorithm and the training data used for the ML. If the